

TechnologyStack

Date	14 July 2026
Team ID	
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap, JavaScript	Provides a responsive and user-friendly interface for entering soil parameters and displaying crop recommendations.
2	Backend / Server-Side	Python, Flask, Scikit-learn	Handles user requests, processes input data, loads the trained Random Forest model, and generates crop predictions.
3	Data Storage	CSV Dataset, Joblib	Stores the crop dataset and trained machine learning model for prediction.
4	Deployment	Flask Development Server	Hosts the web application and enables real-time crop recommendation through a browser.
5	Version Control	Git, GitHub	Maintains source code, tracks project changes, and supports collaboration.
6	Libraries / Tools	Pandas, NumPy, Matplotlib, Joblib	Used for data preprocessing, analysis, visualization, and model serialization.